Healthcare Data Analysis & Disease Risk Prediction

Interactive dashboard, disease risk prediction and health insights



Problem Statement & Objectives



Prepare the data

Clean patient records and prepare useful health indicators

Find patterns

Explore age, BMI, glucose, blood pressure and lifestyle factors

Visualize insights

Charts and filters make health trends easier to understand

Predict disease risk

A machine-learning model estimates risk from selected health indicators

Healthcare Dataset Overview

Field	Type	Example
Patient_ID	Text	P-1042
Record_Date	Datetime	2024-03-14
Age_Group	Category	40–49
BMI, Glucose	Float	28.4, 126
Blood_Pressure	Category	128/82
Lifestyle	Category	Low activity
Family_History	Category	Family history
Diagnosis	Category	Positive / Negative
Age	Integer	45

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Project Workflow



Collect

Load patient health records from a CSV dataset



Preprocess

Handle missing values, duplicates and inconsistent data



Explore

Explore health indicators using summaries and EDA



Visualize

Dashboard, risk charts and disease distribution plots



Interact

Filters, risk prediction and CSV download

Healthcare Data Preprocessing Steps

1. Load & inspect

Check dataset shape, data types, summary statistics and missing values using info() and describe().

2. Missing values

Fill categorical values with the mode and numeric values with the median; review records missing key health measurements.

3. Remove duplicates

Remove repeated patient records using the available patient or record ID.

4. Types & features

Convert columns to suitable types and create useful features such as age groups and BMI categories when available.

5. Standardize labels

Trim spaces, standardize category labels and make diagnosis values consistent.

6. Validate & outliers

Check valid ranges for age, BMI, glucose and blood pressure; investigate outliers and encode diagnosis labels.

Preprocessing in Python (Pandas)



Why it matters

- Clean data supports more reliable analysis
- Age and health indicators drive the analysis
- Valid measurements help avoid misleading results
- One cleaned CSV is reused by every feature

Raw Data vs Clean Data

Raw data

- Duplicate patient records
- Mixed labels in categories and diagnosis
- Missing age or health measurements
- Numbers and categories stored in inconsistent types



Clean data

Saved as healthcare_clean.csv and loaded by the dashboard, prediction model and filters.

Key Features



Healthcare Analysis Dashboard

Explore patient data through charts for age, BMI, glucose, blood pressure and diagnosis.



Disease Risk Prediction

Enter selected health indicators to view an estimated disease-risk category.



Health Indicator Visualizations

Compare health indicators across patient groups using interactive charts.



Disease Risk Analysis

Compare disease and non-disease groups across selected health indicators.



Interactive Filters

Filter patient records by age group, gender and diagnosis where available.



CSV Download

Download the filtered or cleaned healthcare data as a CSV file.

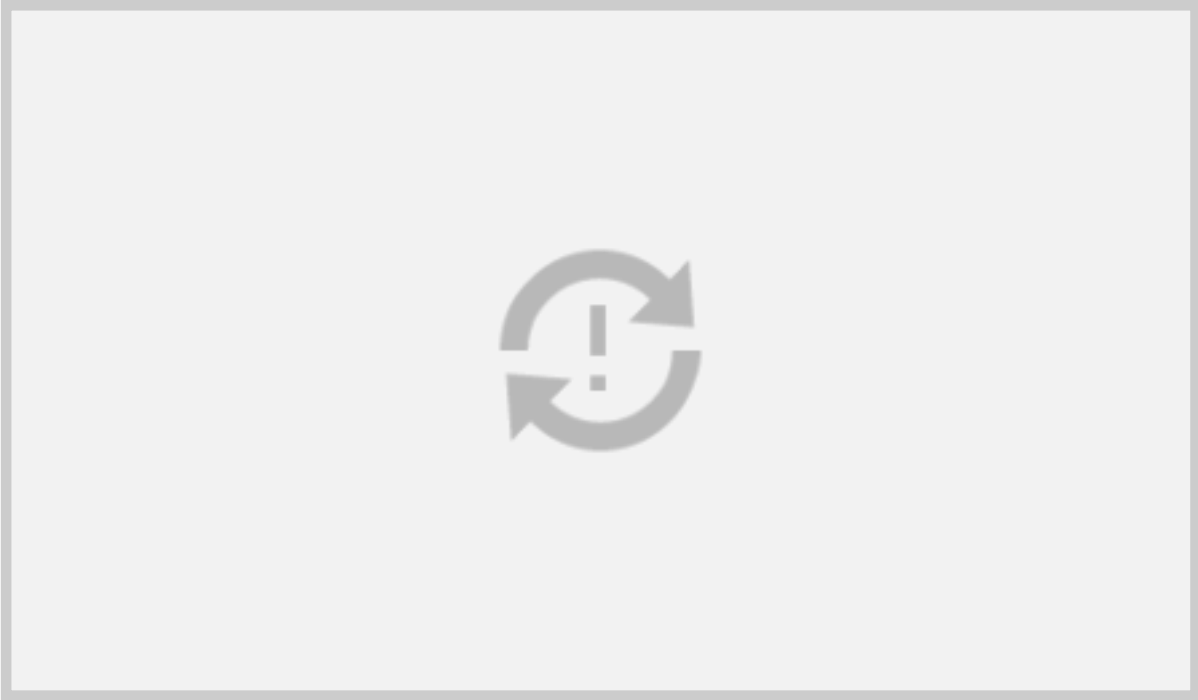
Healthcare Analysis Dashboard

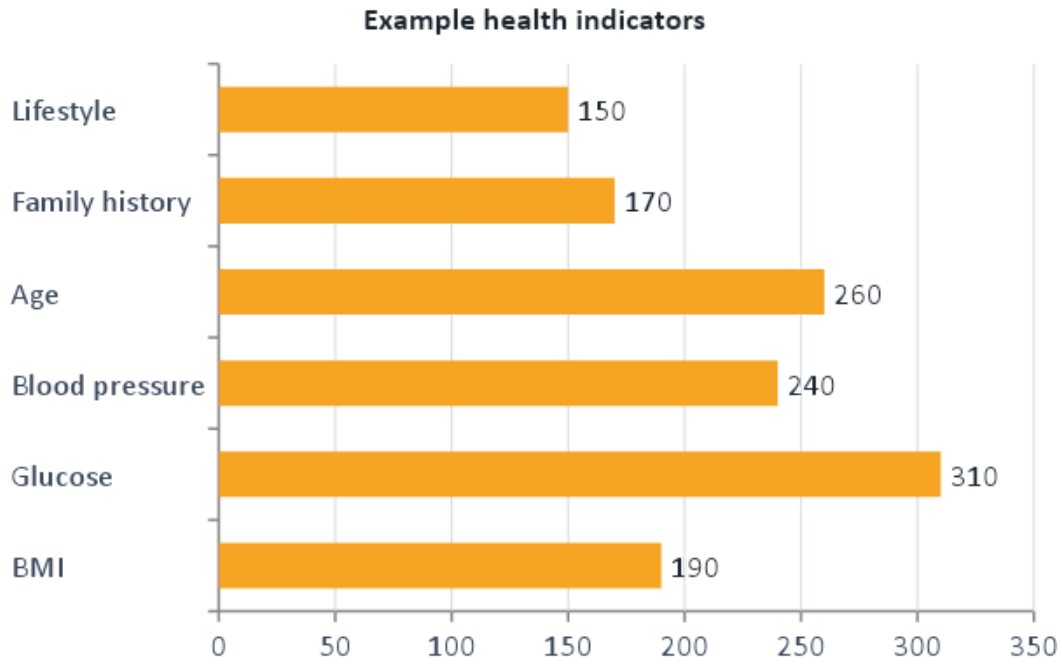
1,250
Total patients

50–69
Peak hours

Positive
Key indicator

18–29
Higher-risk age group





Illustrative sample data

Disease Risk Prediction

Patient health indicators

Glucose level

Enter value

BMI

Enter value

Age

Enter age

67

out of 100

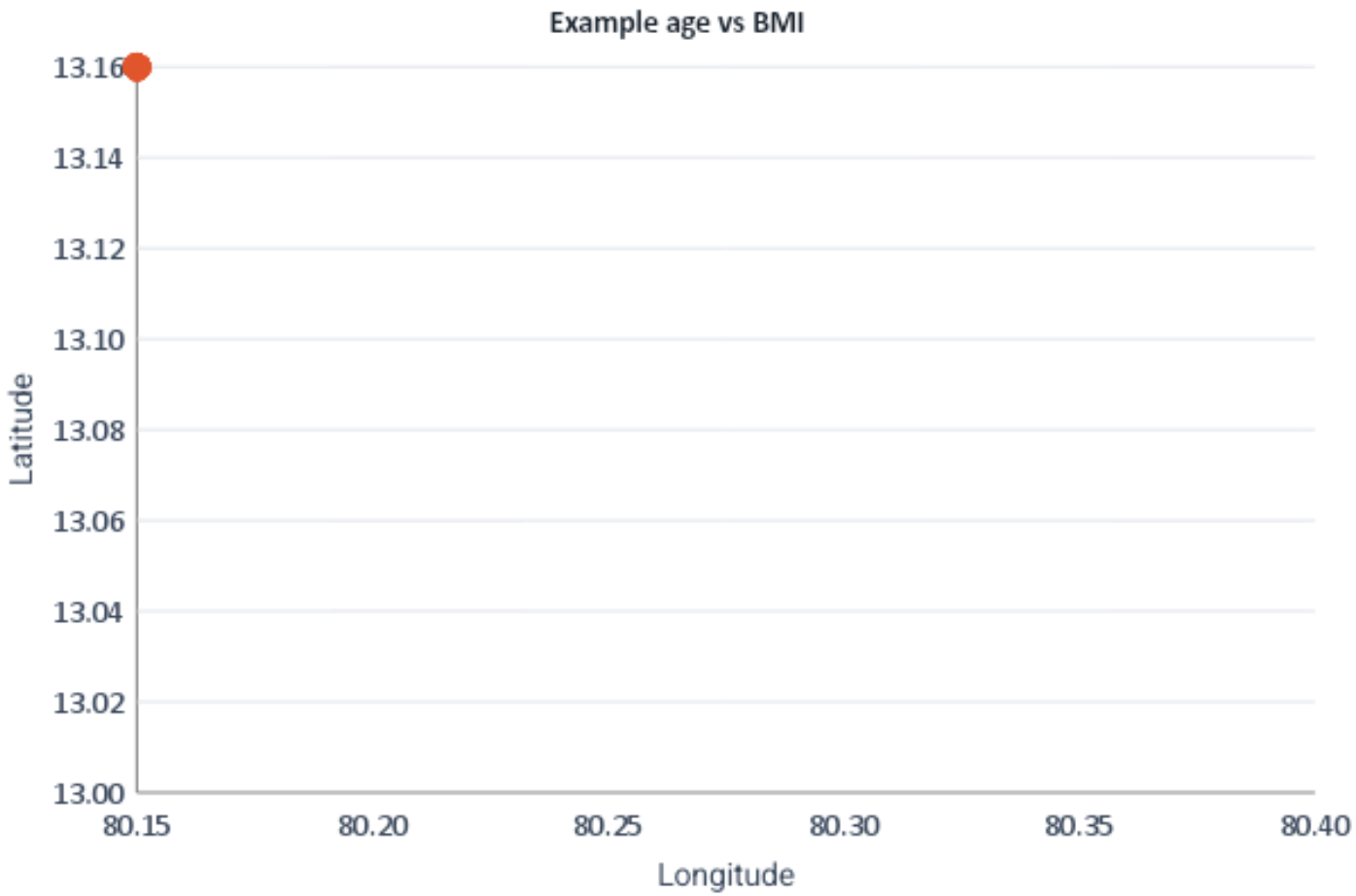
Higher estimated risk

Example prediction inputs

Factor	Points
Base score	20
Weather (Clear/Rain/Fog)	0 / +15 / +20
Road (Dry/Wet/Damaged)	0 / +12 / +18
Time (Day/Evening/Night)	0 / +10 / +20

Example: age, BMI, glucose and blood pressure are passed to the trained model. Risk categories depend on the trained model and dataset. This is an educational prototype, not a medical diagnosis.

Health Indicator Visualizations



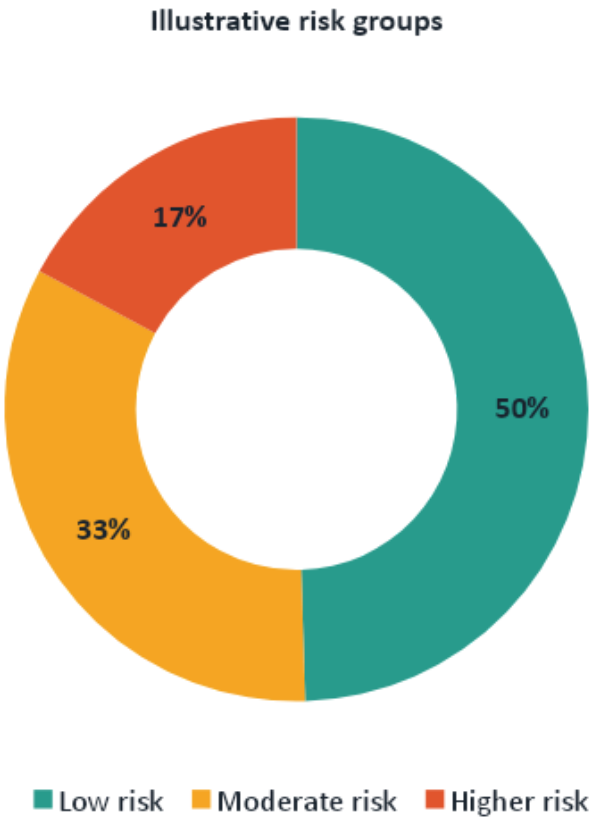
Example patient health measurements

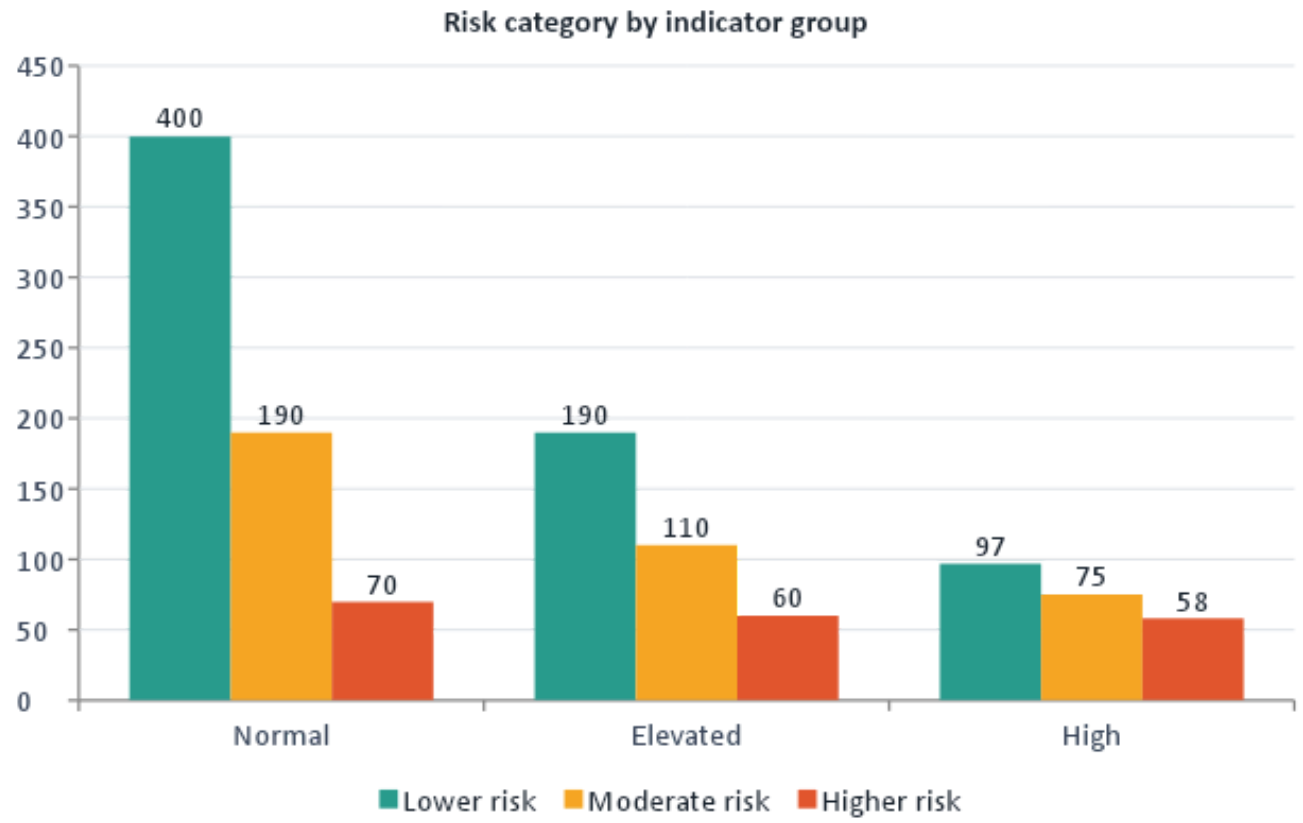
Analysis views

- Compare indicators by diagnosis
- Heatmap of correlations between health indicators
- Group patients into meaningful risk patterns
- Tooltips show health measurements and diagnosis

Built with Plotly using patient health indicators and diagnosis columns.

Disease Risk Analysis





Interactive Filters & CSV Download

Interactive filters

Age group

18–29 | 30–49 | 50+

Diagnosis

Positive | Negative

Risk category

Low | Moderate | High

CSV download

Age_Group	Family_History	Diagnosis
		Serious
		Fatal
	Distracted	Serious

Download CSV

Suggested Tech Stack

Python	Core language
Pandas	Preprocessing and analysis
Plotly	Interactive charts
Folium	Health data visualizations
Streamlit	Web app, filters and download



Conclusion & Future Scope

What we achieved

- Cleaned patient records using a repeatable preprocessing pipeline
- Created dashboard and health-indicator visualizations
- Added filters, risk prediction and CSV export

Future scope

- Train and evaluate machine-learning models for disease-risk prediction
- Integrate verified healthcare datasets or approved health APIs
- Add explainable predictions and a user-friendly interface